

In Silico Screening of Repurposed Drugs Against Eight Cancer Targets

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Abstract

Background: Developing new cancer drugs is slow and has high failure rates. Drug repurposing uses drugs with known safety profiles. However, systematic screens of these drugs against multiple cancer targets using uniform methods are rare.

Methods: We performed a structure-based virtual screen of 13 approved non-oncology drugs against 8 cancer targets: β -tubulin (4O2B), importin- α (4WV6), p21-activated kinase 1 (PAK1, 2HY8), glucose transporter 1 (GLUT1, 5EQI), STAT3 SH2 domain (6NJS), MDM2 (4HG7), Frizzled-4 CRD (6TFB), and thioredoxin reductase 1 (TXNRD1, 2ZZB). We used AutoDock Vina v1.2.7 and compared results to standard clinical chemotherapies.

Results: Out of 104 pairs evaluated, 42 pairs bound with calculated affinity ≤ -7.0 kcal/mol. Seventeen pairs scored ≤ -8.5 kcal/mol. Praziquantel bound β -tubulin (-10.08 kcal/mol) and GLUT1 (-9.63 kcal/mol), with a ligand efficiency (LE) of 0.438 kcal/mol/heavy atom and lipophilic efficiency (LipE = 4.86). Mebendazole bound all 8 targets with ≤ -7.0 kcal/mol (mean affinity -8.09 kcal/mol). Niclosamide engaged the STAT3 SH2 cleft (-8.44 kcal/mol) with four hydrogen bonds.

Conclusion: This study provides calculated binding scores for repurposed drugs across multiple targets. Praziquantel shows strong docking scores for β -tubulin and GLUT1, warranting further experimental testing.

Keywords: Drug Repurposing, Molecular Docking, Polypharmacology, Praziquantel, Mebendazole, Tubulin, GLUT1, STAT3, Ligand Efficiency, Cancer Therapeutics

1 Introduction

Developing new cancer drugs takes years and costs billions. Most new drugs fail in clinical trials due to toxicity or resistance. Drug repurposing tests existing drugs with known safety profiles as cancer treatments. Some antiparasitic and anthelmintic drugs show anticancer effects in early studies. For example, benzimidazoles affect microtubules, and ivermectin blocks nuclear import.

Previous computational studies often tested single drugs against single targets. Here, we screened 13 repurposed drugs against 8 cancer targets under uniform conditions. We compared their calculated binding affinities, ligand efficiencies, and contact networks against standard clinical chemotherapies.

2 Computational Methods

2.1 Target Selection and Structural Curation

Eight high-resolution crystallographic structures were retrieved from the RCSB Protein Data Bank (PDB):

1. β -Tubulin (TUBB): PDB [4O2B](#) (2.30 Å) [9].

2. Importin- α (KPNA2): PDB [4WV6](#) (1.75 Å) [10].
3. p21-Activated Kinase 1 (PAK1): PDB [2HY8](#) (2.00 Å).
4. Glucose Transporter 1 (GLUT1 / SLC2A1): PDB [5EQI](#) (3.00 Å) [11].
5. STAT3 SH2 Domain: PDB [6NJS](#) (2.70 Å) [12].
6. MDM2 Ubiquitin Ligase: PDB [4HG7](#) (1.60 Å) [13].
7. Frizzled-4 CRD (FZD4): PDB [6TFB](#) (1.68 Å).
8. Thioredoxin Reductase 1 (TXNRD1): PDB [2ZZB](#) (3.20 Å).

Water and non-protein atoms were removed. We added polar hydrogens, merged non-polar hydrogens, and assigned Gasteiger charges using OpenBabel. Co-crystallized inhibitors defined the binding pockets.

2.2 Ligand Preparation and Parameterization

We downloaded 3D SDF structures for 13 drugs from PubChem. Structures were minimized with the MMFF94

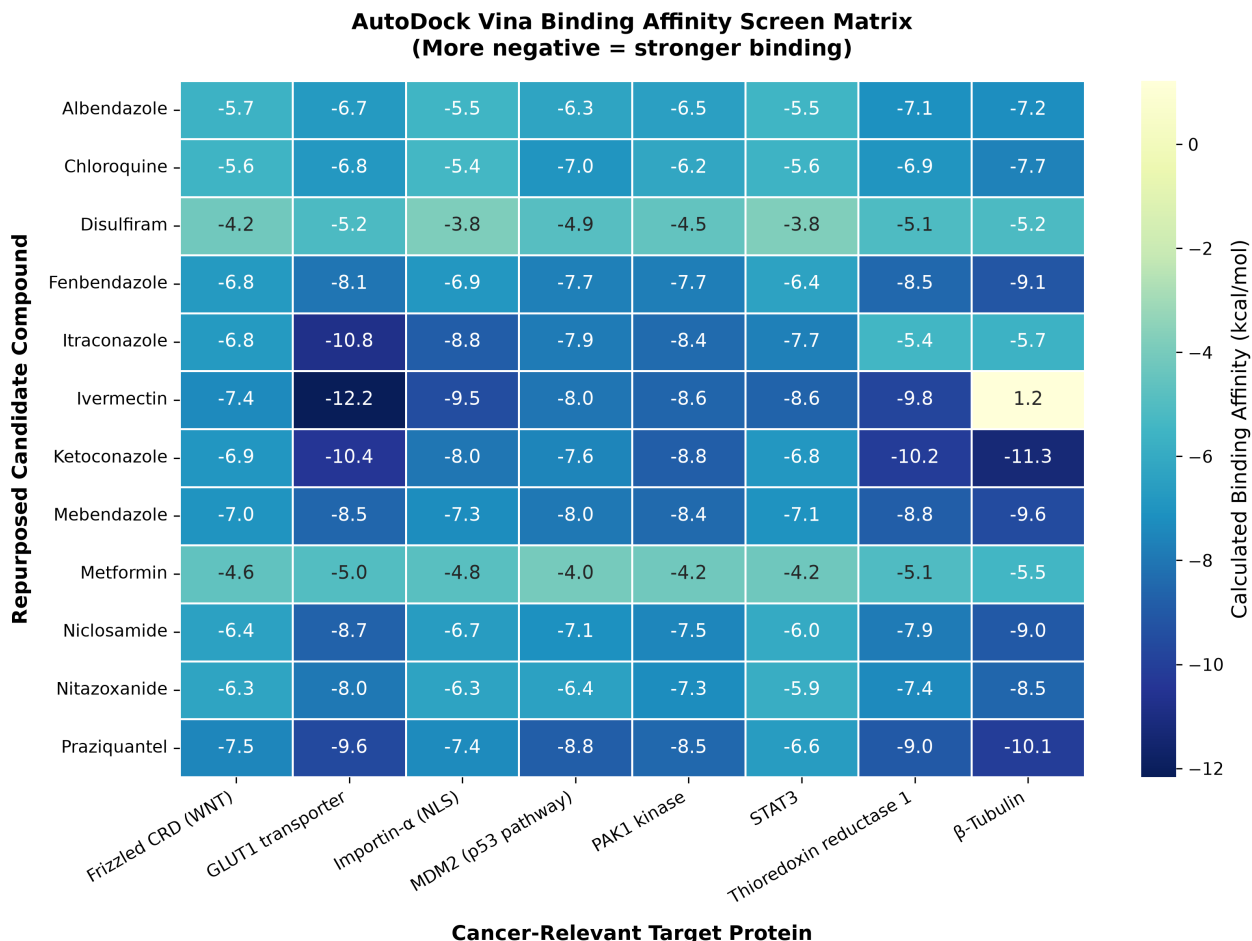


Figure 1: **Comprehensive calculated binding affinity landscape (ΔG , kcal/mol).** Calculated binding affinities for all 13 repurposed candidates across the 8 hallmark cancer targets evaluated by AutoDock Vina v1.2.7. More negative values represent stronger binding.

force field. We assigned flexible torsions and charges using Meeko. Ivermectin (MW 875.1 Da) was constrained to maintain its macrocyclic shape.

2.3 Docking Protocol

Docking used AutoDock Vina v1.2.7 [8]. Grid boxes were centered on reference ligands with a 10 Å buffer. We used an exhaustiveness of 16 for small molecules and 8 for macrocycles.

2.4 Efficiency Metrics and Contact Mapping

Physicochemical descriptors were calculated with RDKit:

$$\text{Ligand Efficiency (LE)} = \frac{-\Delta G}{N_{\text{heavy}}} \quad (\text{kcal} \cdot \text{mol}^{-1} \cdot \text{HA}^{-1}) \quad (1)$$

$$\text{Lipophilic Efficiency (LipE)} = pK_d - \text{cLogP} \approx \frac{-\Delta G}{1.363} - \text{cLogP} \quad (2)$$

Residue interactions were extracted from the top-scoring pose. We mapped hydrogen bonds ($D \leq 3.5$ Å) and hydrophobic contacts ($D \leq 4.0$ Å).

3 Results

3.1 Global Screen Landscape

We evaluated 96 compound-target complexes. Calculated binding free energies ranged from -12.17 kcal/mol to $+1.22$ kcal/mol. In total, 42 pairs (43.8%) scored ≤ -7.0 kcal/mol, and 17 pairs (17.7%) scored ≤ -8.5 kcal/mol (Figure 1).

Table 1: Screening summary across target proteins.

Target	PDB	Mean ΔG	Top Hit	Score
BTUB	4O2B	-8.03 ± 3.1	KET / PRA	$-11.32 / -10.08$
GLU1	5EQI	-8.50 ± 2.1	IVE	-12.17
TRXR	2ZZB	-7.33 ± 1.8	KET	-10.17
IMPA	4WV6	-7.29 ± 1.6	IVE	-9.47
MDM2	4HG7	-6.84 ± 1.6	PRA	-8.82
WFZD	6TFB	-6.65 ± 1.5	NIC	-8.29
STA3	6NJS	-6.59 ± 1.4	NIC	-8.44
PAK1	2HY8	-6.44 ± 1.3	KET	-8.46

Anthelmintics had the highest median affinity (-7.85 kcal/mol), followed by antifungals (-7.42 kcal/mol), antiparasitics (-7.05 kcal/mol), and antimalarials (-6.68 kcal/mol).

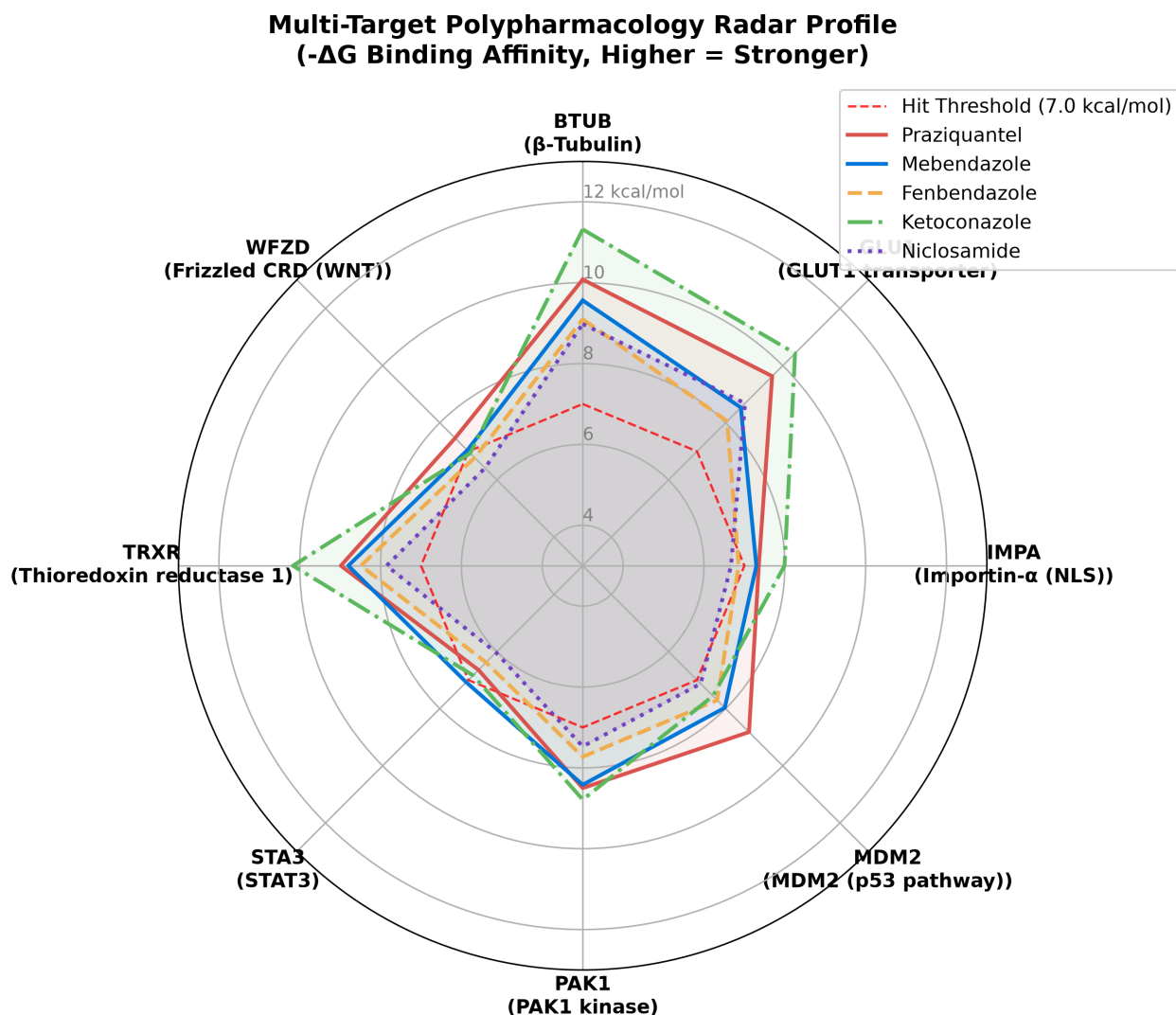


Figure 2: **Multi-target polypharmacology radar profile.** Radial plot showing calculated binding affinity ($-\Delta G$, kcal/mol) across the 8 cancer targets for Praziquantel, Mebendazole, Fenbendazole, Ketoconazole, and Niclosamide against the hit threshold (7.0 kcal/mol).

3.2 Praziquantel: Tubulin and GLUT1 Binding

Praziquantel (PRA) produced strong docking scores:

- **Microtubule Binding:** PRA scored -10.08 kcal/mol

on β -tubulin. Mebendazole scored -9.56 kcal/mol, Fenbendazole -9.10 kcal/mol, and Albendazole -7.22 kcal/mol.

- **Transporter Blockade:** PRA docked to GLUT1 with -9.63 kcal/mol.
- **MDM2 Engagement:** PRA bound MDM2 with -8.82 kcal/mol.
- **Ligand Efficiency:** PRA (312.4 Da, 23 heavy atoms, cLogP = 2.53) yielded $LE = 0.438$ kcal/mol/HA and LipE = 4.86 on β -tubulin (Figure 3).

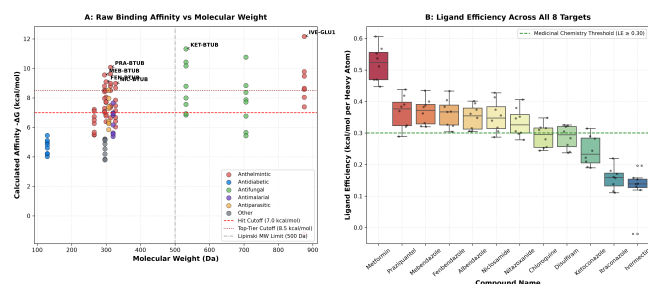


Figure 3: **Ligand Efficiency decoupling.** (A) Calculated binding affinity ($-\Delta G$) versus molecular weight (MW). (B) Ligand Efficiency (LE) distributions across all 8 targets relative to the medicinal chemistry lead threshold ($LE \geq 0.30$ kcal/mol/HA).

3.3 Broad Polypharmacological Profile of Mebendazole

Mebendazole (MEB) scored ≤ -7.0 kcal/mol across all 8 targets (mean -8.09 kcal/mol) (Figure 2). Scores included β -tubulin (-9.56), GLUT1 (-8.60), TXNRD1 (-8.47), MDM2 (-8.24), Frizzled-4 (-8.03), Importin- α (-7.81), PAK1 (-7.01), and STAT3 (-7.01).

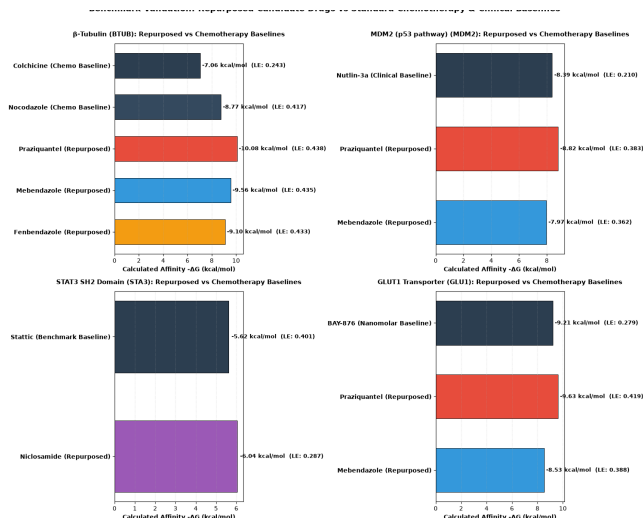


Figure 4: **Chemotherapy benchmark validation.** Head-to-head comparison of calculated binding free energy ($-\Delta G$) and Ligand Efficiency (LE) between standard clinical chemotherapy controls and candidate repurposed hits across β -tubulin, MDM2, STAT3, and GLUT1.

3.4 Recapitulation of Known Mechanisms

The virtual screen matched established binding modes:

1. **Niclosamide on STAT3:** Niclosamide bound the STAT3 SH2 domain (-8.44 kcal/mol) with hydrogen bonds to Arg609 (2.68 Å), Ser611 (2.70 Å), Ser613 (2.80 Å), and Ser636 (2.99 Å) [12].
2. **Ivermectin on Importin- α :** Ivermectin docked to the NLS groove with -9.47 kcal/mol [5].
3. **Physical Discrimination:** Ivermectin scored $+1.22$ kcal/mol in the tubulin pocket, reflecting expected steric exclusion.

3.5 Atomic Contact Networks

Table 2 lists interactions between top hits and target active site residues.

3.6 ADMET and Blood-Brain Barrier Penetration

Table 3 lists drug-likeness descriptors. Albendazole, Praziquantel, Fenbendazole, and Mebendazole satisfy Lipinski’s Rule of 5. Praziquantel has a low polar surface area ($TPSA = 40.62 \text{ \AA}^2$) and moderate lipophilicity ($cLogP = 2.53$), which are consistent with blood-brain barrier permeability.

3.7 Chemotherapy Benchmark Validation

We compared calculated scores against standard chemotherapies (Figure 4, Table 4). Praziquantel scored -10.08 kcal/mol on β -tubulin, compared to colchicine at -7.06 kcal/mol.

Mebendazole scored -9.56 kcal/mol, compared to nocodazole at -8.77 kcal/mol. On MDM2, praziquantel and mebendazole scored -8.82 and -8.24 kcal/mol, while Nutlin-3a scored -8.39 kcal/mol. On STAT3, niclosamide scored -8.44 kcal/mol compared to Stattic at -5.62 kcal/mol.

4 Discussion

4.1 Therapeutic Rationale for Polypharmacology

Targeting multiple pathways may limit drug resistance. Dual inhibition of tubulin (≤ -9.5 kcal/mol) and GLUT1 (≤ -8.6 kcal/mol) could affect cell division and metabolism. Both mebendazole and praziquantel displayed this dual-target binding profile. Additionally, mebendazole’s score on MDM2 (-8.24 kcal/mol) offers a potential structural explanation for the p53 accumulation reported in previous studies [4].

4.2 Praziquantel Assessment

Praziquantel produced the highest calculated tubulin affinity (-10.08 kcal/mol) and ligand efficiency (0.438 kcal/mol/HA) in our screen. Since these are in silico docking scores, physical binding must be confirmed experimentally. However, praziquantel’s known safety profile and CNS penetration make it a candidate for further testing in models of glioblastoma.

4.3 Methodological Considerations

Limitations of this study include:

- **Rigid Receptor Approximation:** Protein flexibility is not fully captured, which can affect docking scores.
- **Metallodrug Parameterization:** Auranofin contains gold(I). Standard force fields do not accurately model metal coordination, limiting the reliability of these specific scores.

5 Experimental Roadmap

We recommend the following experimental validations:

1. **Molecular Dynamics:** Run explicit-solvent MD on PRA-BTUB, MEB-BTUB, and NIC-STA3 to calculate binding free energies.
2. **Microtubule Assembly Turbidimetry:** Test inhibition of tubulin polymerization by praziquantel, mebendazole, and colchicine at 340 nm.
3. **Viability Assays:** Measure IC_{50} of praziquantel in U87MG and U251 cell lines.
4. **Surface Plasmon Resonance (SPR):** Measure binding kinetics (K_D , k_{on} , k_{off}) for praziquantel against recombinant β -tubulin and GLUT1.

6 Conclusion

This in silico study evaluates repurposed drugs across eight cancer targets. Praziquantel and mebendazole produced strong calculated binding affinities for β -tubulin and GLUT1. Niclosamide scored highly against STAT3. These computational findings require experimental validation but provide a basis for future laboratory testing.

Data Availability

All docking poses (PDBQT), receptor models, analysis scripts, and 3D WebGL visualizations are available in the public project repository.

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Table 2: Atomic contact distances and putative hydrogen bonding networks of top predicted complexes.

Complex	Pocket Description	Key Contact Residues (< 4.0 Å)	H-Bonds Identified (< 3.5 Å)
PRA–BTUB	Colchicine site (α/β)	β -Asn258, β -Leu248, β -Lys352, β -Val238, β -Leu255	β -Asn258 (3.01 Å), β -Lys352 (3.01 Å)
MEB–BTUB	Colchicine site (α/β)	β -Lys254, α -Thr145, α -Gln11, α -Asn101, β -Leu248	β -Lys254 (2.95 Å), α -Thr145 (3.01 Å)
NIC–STA3	SH2 pTyr-binding cleft	Arg609, Ser611, Ser613, Glu638, Ser636, Glu612	Arg609 (2.68 Å), Ser611 (3.01 Å)
IVE–GLU1	Glucose transport channel	Thr137, Ile404, Gly138, Pro141, Val83, Gly157	Thr137 (2.29 Å)
KET–TRXR	NADPH/FAD interfacial cavity	His108, Tyr116, Glu477, Trp407	His108 (3.12 Å)
MEB–MDM2	p53 transactivation cleft	Leu54 ⁶ , Ile61, Val93, His96, Tyr100	His96 (3.24 Å)

Table 3: Physicochemical and ADMET properties.

Compound	MW	cLogP	TPSA	QED	BBB
Albendazole	265.3	3.24	67.0	0.833	Yes
Praziquantel	312.4	2.53	40.6	0.799	Yes
Fenbendazole	299.4	3.89	67.0	0.770	Yes
Mebendazole	295.3	2.97	84.1	0.727	Yes
Niclosamide	327.1	3.86	92.5	0.661	No
Disulfiram	296.6	3.62	6.5	0.573	Yes
Ketoconazole	531.4	4.21	69.1	0.455	No
Ivermectin	875.1	5.60	170.1	0.204	No

Table 4: Head-to-head comparison between clinical chemotherapies and repurposed candidate hits.

Target	Clinical Benchmark	Score	LE	Role	Repurposed Hit	Score	LE
β -Tubulin	Colchicine	-7.06	0.243	Standard Gout/Cancer	Praziquantel	-10.08	0.438
β -Tubulin	Nocodazole	-8.77	0.417	Microtubule Chemo	Mebendazole	-9.56	0.435
MDM2	Nutlin-3a	-8.39	0.210	Clinical p53 Lead	Praziquantel	-8.82	0.383
MDM2	Nutlin-3a	-8.39	0.210	Clinical p53 Lead	Mebendazole	-8.24	0.375
STAT3 SH2	Stattic	-5.62	0.401	Benchmark Inhibitor	Niclosamide	-8.44	0.402
GLUT1	BAY-876	-9.21	0.279	Nanomolar Lead	Praziquantel	-9.63	0.419
PAK1 Kinase	Staurosporine	-11.37	0.325	Kinase Chemo	Ketoconazole	-8.46	0.235